

2.6 - Inverse of a Function

Inverse of a function:

- The inverse of a function f is denoted as f^{-1}
- The function and its inverse have the property that if $f(a) = b$, then $f^{-1}(b) = a$
- So if $f(5) = 13$, then $f^{-1}(13) = 5$.
- More simply put: The inverse of a function has all the same points as the original function, except that the x 's and y 's have been reversed.

It is important to note that $f^{-1}(x)$ is read as "the inverse of f at x ". The -1 does not behave like an exponent.

$$f^{-1}(x) \neq \frac{1}{f(x)}$$



To draw an inverse, all you need to do is swap the x and y coordinates of each point.



Finding Inverses by Numerically

Example 1: The table shows ordered pairs belonging to a function $f(x)$. Determine $f^{-1}(x)$, then state the domain and range of $f(x)$ and its inverse.

$f(x)$	$f^{-1}(x)$
$(-5, 0)$	$(0, -5)$
$(-4, 2)$	$(2, -4)$
$(-3, 5)$	$(5, -3)$
$(-2, 6)$	$(6, -2)$
$(0, 7)$	$(7, 0)$

$f(x)$

$$D: \{x \in \mathbb{R} \mid x = -5, -4, -3, -2, 0\}$$

$$R: \{y \in \mathbb{R} \mid y = 0, 2, 5, 6, 7\}$$

$f^{-1}(x)$

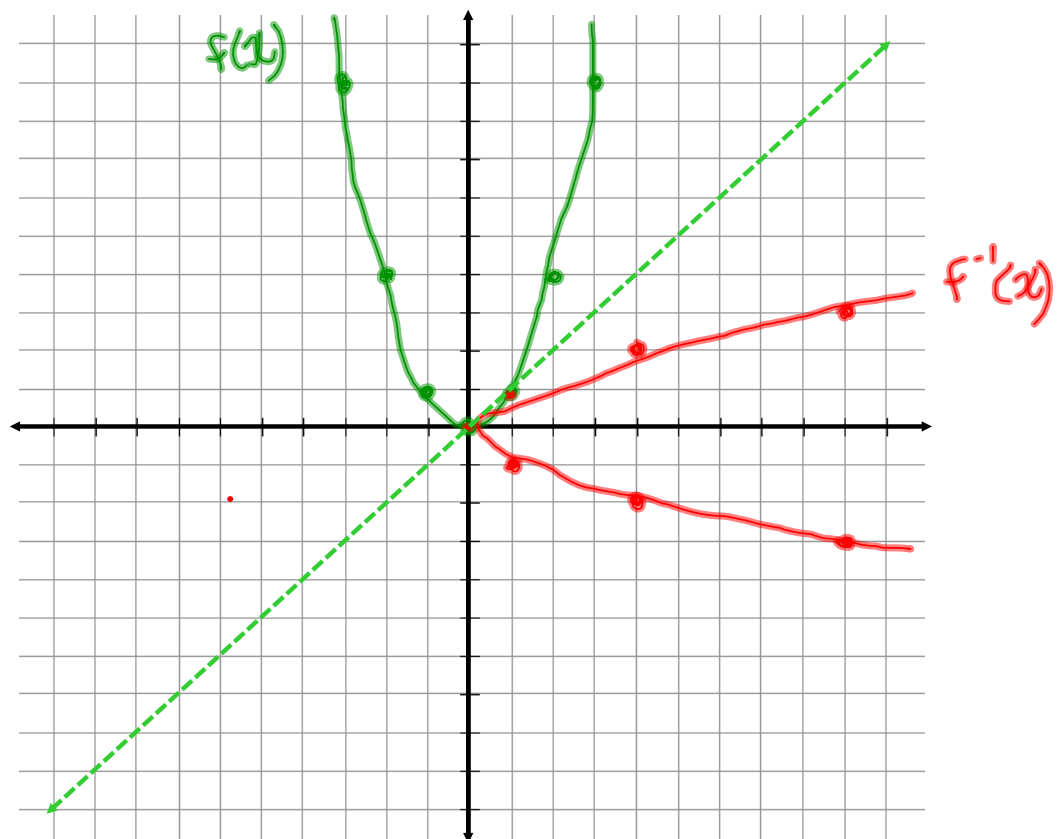
$$D: \{x \in \mathbb{R} \mid x = 0, 2, 5, 6, 7\}$$

$$R: \{y \in \mathbb{R} \mid y = -5, -4, -3, -2, 0\}$$

Example 2:

a) Graph the function $f(x) = x^2$ and its inverse $f^{-1}(x)$

$f(x)$	$f^{-1}(x)$
$(-3, 9)$	$(9, -3)$
$(-2, 4)$	$(4, -2)$
$(-1, 1)$	$(1, -1)$
$(0, 0)$	$(0, 0)$
$(1, 1)$	$(1, 1)$
$(2, 4)$	$(4, 2)$
$(3, 9)$	$(9, 3)$



b) state the domain and range of both functions

$$\underline{f(x)}$$

$$D: \{x \in \mathbb{R}\}$$

$$R: \{y \in \mathbb{R} \mid y \geq 0\}$$

$$\underline{f^{-1}(x)}$$

$$D: \{x \in \mathbb{R} \mid x \geq 0\}$$

$$R: \{y \in \mathbb{R}\}$$

Note: the domain and range of inverse functions are the reverse of each other.

Example 3:

Sketch the graph of $g(x) = -2\sqrt{-\frac{1}{2}x} + 3$ then graph $g^{-1}(x)$.

$$f(x) = \sqrt{x} \rightarrow$$

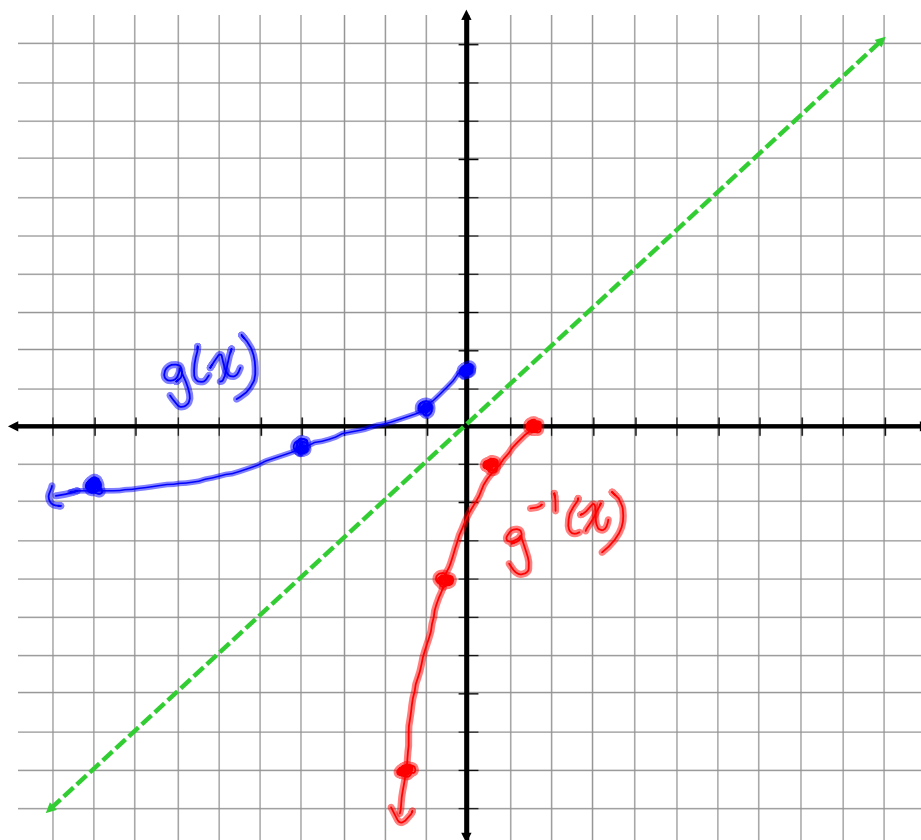
x	y
0	0
1	1
4	2
9	3

$$g(x) = -2\sqrt{-\frac{1}{2}x} + 3 \rightarrow$$

$-2x$	$-2y+3$
0	3
-2	1
-8	-1
-18	-3

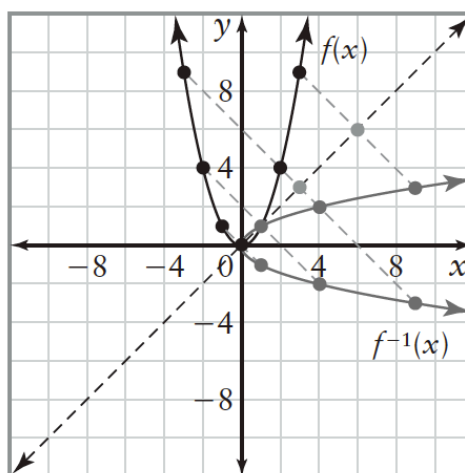
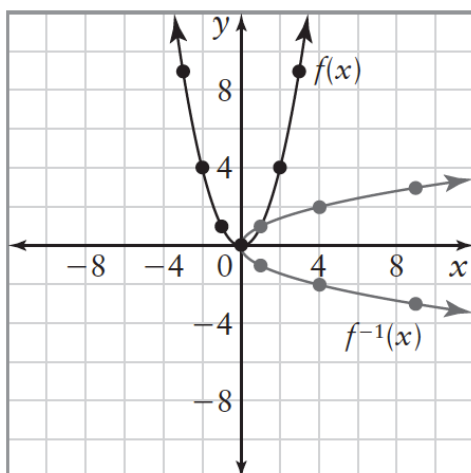
$$g^{-1}(x)$$

x	y
3	0
1	-2
-1	-8
-3	-18

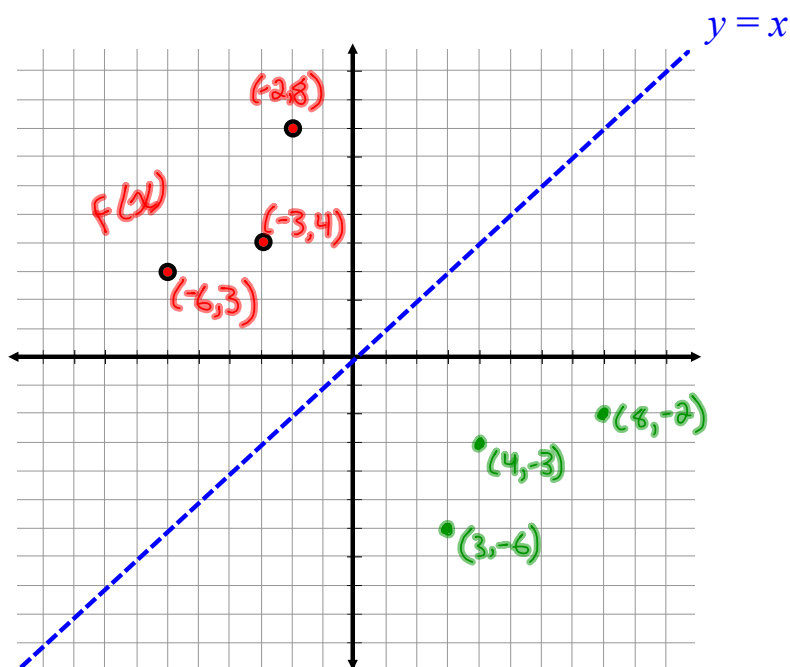


Finding Inverses by Graphing

The graph of $f^{-1}(x)$ is the graph of $f(x)$ reflected in the line $y = x$. This is true for all functions and their inverses. If you find the midpoint of each pair of points from example 2 and connect them you can prove this theorem.



Example 4: Sketch the inverse of the $f(x)$



Finding Inverses Algebraically

Algebraic Method for finding the inverse:

1. Replace $f(x)$ with "y"
2. Switch the x and y variables
3. Isolate for y
4. replace y with $f^{-1}(x)$

Example 5: Find the inverse of the following functions...

a) $g(x) = \frac{(3x)}{4}$

$$y = \frac{3x}{4}$$

$$x = \frac{3y}{4}$$

$$4x = 3y$$

$$\frac{4x}{3} = y$$

$$g^{-1}(x) = \frac{4x}{3}$$

b) $h(x) = 4x + 3$

$$y = 4x + 3$$

$$x = 4y + 3$$

$$x - 3 = 4y$$

$$\frac{x-3}{4} = y$$

$$h^{-1}(x) = \frac{x-3}{4}$$

c) $f(x) = x^2 - 1$

$$y = x^2 - 1$$

$$x = y^2 - 1$$

$$x + 1 = y^2$$

$$\pm\sqrt{x+1} = y$$

$$f^{-1}(x) = \pm\sqrt{x+1}$$

d) $h(x) = \frac{4x+3}{5}$

$$y = \frac{4x+3}{5}$$

$$x = \frac{4y+3}{5}$$

$$5x = 4y + 3$$

$$\frac{5x-3}{4} = y$$

$$h^{-1}(x) = \frac{5x-3}{4}$$

e) $f(x) = 2x^2 + 16x + 29$

$$y = (2x^2 + 16x) + 29$$

$$y = 2(x^2 + 8x + 16 - 16) + 29$$

$$y = 2(x^2 + 8x + 16) - 32 + 29$$

$$y = 2(x+4)^2 - 3$$

$$x = 2(y+4)^2 - 3$$

$$\frac{x+3}{2} = (y+4)^2$$

$$\pm \sqrt{\frac{x+3}{2}} = y+4$$

$$-4 \pm \sqrt{\frac{x+3}{2}} = y$$

$$f^{-1}(x) = -4 \pm \sqrt{\frac{x+3}{2}}$$

Note: for algebraic inverses of quadratic functions, before interchanging x and y 's you must re-write in vertex form.

f) $r(x) = \sqrt{x} + 2$

$$y = \sqrt{x} + 2$$

$$x = \sqrt{y} + 2$$

$$x - 2 = \sqrt{y}$$

$$(x-2)^2 = y$$

$$r^{-1}(x) = (x-2)^2$$